

GARIMA, AMRSC

Computational & Theoretical Chemist · PhD, IIT Roorkee · DFT · MD · CADD

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PROFESSIONAL SUMMARY

Computational and theoretical chemist with a PhD (thesis submitted, 2026) from IIT Roorkee, specializing in molecular dynamics, Monte Carlo, Langevin/Brownian dynamics, density functional theory (DFT), and computer-aided drug design (CADD). Skilled in stochastic and coarse-grained simulation, electronic-structure calculations, protein–ligand docking, virtual screening, and scientific programming (C, Fortran, Python, Mathematica, HPC), with a strong analytical and spectroscopy foundation (HPLC, NMR, FTIR, UV-Vis).

EDUCATION

Ph.D., Theoretical & Computational Chemistry Indian Institute of Technology Roorkee <i>CGPA 8.625 / 10</i>	2018 – 2026 (expected)
B.Ed. Chaudhary Ranbir Singh University	2016 – 2018
M.Sc., Chemistry Kurukshetra University <i>81.7% · Top 2% of class</i>	2014 – 2016
B.Sc. Govt. National College (C.D.L.U.), Sirsa <i>85.17% · Top 1% of class</i>	2011 – 2014

QUALIFYING EXAMINATIONS & MEMBERSHIP

- CSIR-UGC NET, Junior Research Fellowship (Chemical Sciences), All India Rank 58, December 2017.
- GATE (Chemistry), All India Rank 341, 2018.
- Associate Member, Royal Society of Chemistry (AMRSC), 2026.

RESEARCH EXPERIENCE

Doctoral Researcher Indian Institute of Technology Roorkee <i>Roorkee</i>	Jul 2018 – Present
- Modeled rupture and rebinding dynamics of bead-spring polymer systems governed by force-dependent catch and slip bond kinetics. - Built stochastic simulations (averaged over 1,000 trajectories) and mean-field formalisms using Fokker–Planck and master equations. - Derived analytical solutions for strongly nonlinear coupled PDEs via the Homotopy Analysis Method (HAM) and traveling-wave formalism. - Implemented numerical solvers in C (4th-order Runge–Kutta with adaptive step size) and Mathematica for rupture-front velocity analysis. - Presented findings at two international conferences; awarded Best Poster twice.	

COMPUTATIONAL TRAINING & PROJECTS

Computer-Aided Drug Design (CADD) Centre for Advanced Computational Research (CACR), New Delhi <i>Delhi</i>	<i>New Delhi</i> Nov – Dec 2025
- Performed protein–ligand molecular docking with AutoDock Vina, including protein/ligand preparation and binding-affinity evaluation. - Applied virtual screening, reverse screening, and ADMET prediction for computational drug discovery; visualized interactions in PyMOL.	
Modelling Quantum Batteries as a Many-Body System (Project) PaAC, IIT Roorkee	Oct 2023 – Jan 2024
- Modelled quantum batteries as interacting many-body quantum systems, implementing numerical simulations in Python and Mathematica. - Computed energy and charging dynamics of the many-body model to analyze quantum battery performance.	

DFT & Computational Materials Physics (Course Project) | Ghent University (online)

Sep – Dec 2023

- Performed periodic DFT calculations on aluminium in Quantum ESPRESSO — geometry optimization, electronic structure, and density of states.
- Analyzed structure–property relationships using the plane-wave DFT workflow.

Scientific Web Development | LLAMPS Lab, Dept. of Chemistry, IIT Roorkee

Jun – Oct 2025

Roorkee

- Built and deployed a React.js research-group website on the IIT Roorkee server; developed a secondary site on Wix.

RESEARCH INTERNSHIP

Summer Research Intern — Synthesis of 2-Substituted Heterocyclic Amines | NIPER, Mohali

Jun – Jul 2015

Mohali

- Synthesized 2-substituted heterocyclic amines under Dr. A. K. Chakraborti.
- Hands-on HPLC, spectroscopy, and column chromatography; reaction setups under inert (N₂) atmosphere; safe chemical handling and detailed documentation.

TECHNICAL SKILLS

Computational chemistry: Molecular dynamics (GROMACS) · DFT (Quantum ESPRESSO, ORCA, Gaussian 09) · Molecular docking (AutoDock Vina) · CADD · Virtual screening · ADMET · Multiwfn · VMD · ChimeraX · CHARMM-GUI · MDAnalysis · RDKit · PyMOL

Programming & numerical: C · Fortran 95 · Python · Mathematica · Xmgrace · HPC clusters · HTML/CSS

Analytical & laboratory: HPLC · NMR · FTIR · UV-Vis spectroscopy · TLC · Column chromatography · Potentiometry · Fluorimetry · Polarimetry · Lab documentation & safety

CONFERENCE PRESENTATIONS & AWARDS

- Best Poster Prize — Theoretical Chemistry Symposium (TCS 2021), IISER Kolkata, Dec 2021.
- Best Poster Award — Research Scholars' Day, Dept. of Chemistry, IIT Roorkee, Nov 2022.
- Poster presentation — CompFlu-2022, IIT Kharagpur, Dec 2022.
- Workshop — Data Acquisition & Analysis in NMR and HR-MS (Accelerate Vigyan, SERB), NIT Calicut, Sep 2021.

CERTIFICATIONS

- Faculty Development Programme (FDP) — Principles and Applications of Quantum Computing, IIT Mandi (Center for Quantum Science & Technologies), June 2026.
- Bioinformatics Specialization (in progress) — UC San Diego, Coursera, 2026.
- Fundamentals of Materials Science — Shanghai Jiao Tong University, Coursera, 2025.
- AI For Everyone — DeepLearning.AI, 2025 · Google Cybersecurity Specialization — Google, 2025 · Python for Beginners.

TEACHING & MENTORING

- Teaching Assistant, IIT Roorkee (2018–2022): supervised UG/PG laboratory sessions; instructed NMR, FTIR, UV-Vis, and TLC techniques.
- Academic Support Specialist (Chemical Sciences), Biotechnika Info Labs (2024–2025): CSIR-NET JRF coaching, live problem-solving, and mock tests.
- Academic Educator & Mentor, Mentor Match / Edvi / Edugraff (2020–present): chemistry tutoring across IGCSE, IB, AP, and JEE curricula.
- Lecturer, Guru Kashi University, Talwandi Sabo (2016–2017): undergraduate General, Organic, Inorganic, and Physical Chemistry.

LANGUAGES

English (C1) · Hindi (Native) · Punjabi (C1)